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Rotating a 3D volume in extended reality

Abstract

An eXtended Reality (XR) system provides methodologies for rotating a virtual object in an XR user interface of the XR system. The methodologies include providing to a user an XR user interface of an XR system, where the XR user interface includes a virtual object displayed to the user. The XR system determines a first and second selection location of a pinch gesture being made by the user and rotates the virtual object using the first and second selection locations and an angle offset of the first selection location. The XR system redisplay the rotated virtual object to the user in the XR user interface.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to user interfaces and, more particularly, to user interfaces used for extended reality.

BACKGROUND

(2) A head-wearable apparatus may be implemented with a transparent or semi-transparent display through which a user of the head-wearable apparatus can view the surrounding environment. Such head-wearable apparatuses enable a user to see through the transparent or semi-transparent display to view the surrounding environment, and to also see objects (e.g., virtual objects such as a rendering of a 2D or 3D graphic model, images, video, text, and so forth) that are generated for display to appear as a part of, and/or overlaid upon, the surrounding environment. This is typically referred to as “augmented reality” or “AR.” A head-wearable apparatus may additionally completely occlude a user's visual field and display a virtual environment through which a user

may move or be moved. This is typically referred to as “virtual reality” or “VR.” In a hybrid form, a view of the surrounding environment is captured using cameras, and then that view is displayed along with augmentation to the user on displays the occlude the user's eyes. As used herein, the term eXtended Reality (XR) refers to augmented reality, virtual reality and any of hybrids of these technologies unless the context indicates otherwise.

(3) A user of the head-wearable apparatus may access and use a computer software application to perform various tasks or engage in an entertaining activity. To use the computer software application, the user interacts with a user interface provided by the head-wearable apparatus.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some non-limiting examples are illustrated in the figures of the accompanying drawings in which:

(2) FIG. 1A is a perspective view of a head-worn device, in accordance with some examples.

(3) FIG. 1B illustrates a further view of the head-worn device of FIG. 1A, in accordance with some examples.

(4) FIG. 2 is a diagrammatic representation of a machine within which a set of instructions may be executed to cause the machine to perform any one or more of the methodologies discussed herein, in accordance with some examples.

(5) FIG. 3A illustrates a collaboration diagram of components of an XR system using hand-tracking for user input, in accordance with some examples.

(6) FIG. 3B illustrates a process flow diagram of a method of using a pinch gesture as user input, in accordance with some examples.

(7) FIG. 3C illustrates using a pinch gesture to rotate a virtual object, in accordance with some examples.

(8) FIG. 3D illustrates a 3D coordinate system, a virtual object, and a rotate collider, in accordance with some examples.

(9) FIG. 3E illustrates a view of a virtual object in a 3D coordinate system having as its origin a virtual object center point of the virtual object, in accordance with some examples.

(10) FIG. 3F illustrates a projection of a virtual object onto a 2D plane defined by a Z axis and an X axis, in accordance with some examples.

(11) FIG. 3G illustrates a translation of a selection location, in accordance with some examples.

(12) FIG. 3H illustrates a rotated virtual object projected onto a 2D plane, in accordance with some examples.

(13) FIG. 3I illustrates a rotated virtual object in an XR user interface, in accordance with some examples.

(14) FIG. 4 illustrates a system of a head-wearable apparatus, in accordance with some examples.

(15) FIG. 5 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

(16) FIG. 6 is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

(17) FIG. 7 is a diagrammatic representation of a messaging system that has both client-side and server-side functionality, in accordance with some examples.

(18) FIG. 8 is a block diagram showing a software architecture, in accordance with some examples.

DETAILED DESCRIPTION

(19) Hand-tracking is a way to provide user inputs from a user into an XR user interface provided by an XR system. The XR system tracks one or more of the user's hands using cameras and computer vision methodologies. The XR system determines hand poses or gestures being made by the user using video images captured by the cameras. In some XR systems, the XR user interface includes one or more virtual objects that are manipulated by the user, termed Direct Manipulation of Virtual Objects (DMVO). Manipulation of the virtual objects may include operations to rotate a virtual object without also moving the object out of its current position.

(20) Use of human input devices that are touched-based or haptic is not an optimal input modality for XR systems. To achieve an immersive or useful user interface, the attention of the user is directed to the virtual objects being displayed to the user and having the user divert their attention to make a selection from a keyboard, keypad, touch surface, or the like is disruptive to the user's attention on the virtual object. In addition, XR systems are designed to be as small as possible so that they are unobtrusive. Accordingly, there is limited surface area available for haptic input devices. Finally, having haptic input devices adds complexity and additional potential points of failure to the XR system. As an XR system may not have haptic inputs or may not be able to track both hands of the user, methodologies for determining the user's intent during a rotating operation on a virtual object using one hand of the user are desirable.

(21) Certain examples of the present disclosure provide methodologies for rotating a virtual object using a selection hand gesture or hand pose such as, but not limited to, a pinch gesture. The user makes a pinch gesture in a location near the virtual object and then moves the location of the pinch gesture. The virtual object is rotated depending on how far the user moves their hand while making the pinch gesture.

(22) In some examples, an XR system provides to a user an XR user interface where the XR user interface includes a virtual object displayed to the user. The XR system determines a first location of a pinch gesture being made by the user in the vicinity of the virtual object and a second location of the pinch gesture and rotates the virtual object using the first location, the second location, and an orientation vector of the virtual object. The XR system redisplay the rotated virtual object to the user in the XR user interface.

(23) In some examples, determining a location of a pinch gesture further includes capturing, using one or more cameras of the XR system, tracking image data. The XR system determines hand-tracking data using the tracking image data and determines the first location of the pinch gesture and the second location of the pinch gesture using the hand-tracking data.

(24) In some examples, to determine a selection location, a rotate collider of the virtual object is created along with a selection location collider. The selection location is determined at an intersection or collision between the selection location collider and the rotate collider.

(25) In some examples, a first selection location vector is generated using the first selection location and a virtual object center point of the virtual object and a second selection location vector is generated using the second selection location and the virtual object center point of the virtual object; and rotating the virtual object using the first selection location vector and the second selection location vector.

(26) In some examples, the XR system detects that the pinch gesture is still being held by the user, and in response to detecting that the pinch gesture is still being held, continues to rotate the virtual object.

(27) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(28) FIG. 1A is a perspective view of a head-wearable apparatus **100** in accordance with some examples. The head-wearable apparatus **100** may be a client device of an XR system, such a computing system **502** of FIG. 5. The head-wearable apparatus **100** can include a frame **102** made from any suitable material such as plastic or metal, including any suitable shape memory alloy. In one or more examples, the frame **102** includes a first or left optical element holder **104** (e.g., a

display or lens holder) and a second or right optical element holder **106** connected by a bridge **112**. A first or left optical element **108** and a second or right optical element **110** can be provided within respective left optical element holder **104** and right optical element holder **106**. The right optical element **110** and the left optical element **108** can be a lens, a display, a display assembly, or a combination of the foregoing. Any suitable display assembly can be provided in the head-wearable apparatus **100**.

(29) The frame **102** additionally includes a left arm or left temple piece **122** and a right arm or right temple piece **124**. In some examples the frame **102** can be formed from a single piece of material so as to have a unitary or integral construction.

(30) The head-wearable apparatus **100** can include a computing device, such as a computer **120**, which can be of any suitable type so as to be carried by the frame **102** and, in one or more examples, of a suitable size and shape, so as to be partially disposed in one of the left temple piece **122** or the right temple piece **124**. The computer **120** can include one or more processors with memory, wireless communication circuitry, and a power source. As discussed below, the computer **120** comprises low-power circuitry **426**, high-speed circuitry **428**, and a display processor. Various other examples may include these elements in different configurations or integrated together in different ways. Additional details of aspects of the computer **120** may be implemented as illustrated by the machine **200** discussed herein.

(31) The computer **120** additionally includes a battery **118** or other suitable portable power supply. In some examples, the battery **118** is disposed in left temple piece **122** and is electrically coupled to the computer **120** disposed in the right temple piece **124**. The head-wearable apparatus **100** can include a connector or port (not shown) suitable for charging the battery **118**, a wireless receiver, transmitter or transceiver (not shown), or a combination of such devices.

(32) The head-wearable apparatus **100** includes a first or left camera **114** and a second or right camera **116**. Although two cameras are depicted, other examples contemplate the use of a single or additional (i.e., more than two) cameras.

(33) In some examples, the head-wearable apparatus **100** includes any number of input sensors or other input/output devices in addition to the left camera **114** and the right camera **116**. Such sensors or input/output devices can additionally include biometric sensors, location sensors, motion sensors, and so forth.

(34) In some examples, the left camera **114** and the right camera **116** provide tracking image data for use by the head-wearable apparatus **100** to extract 3D information from a real-world scene.

(35) The head-wearable apparatus **100** may also include a touchpad **126** mounted to or integrated with one or both of the left temple piece **122** and right temple piece **124**. The touchpad **126** is generally vertically-arranged, approximately parallel to a user's temple in some examples. As used herein, generally vertically aligned means that the touchpad is more vertical than horizontal, although potentially more vertical than that. Additional user input may be provided by one or more buttons **128**, which in the illustrated examples are provided on the outer upper edges of the left optical element holder **104** and right optical element holder **106**. The one or more touchpads **126** and buttons **128** provide a means whereby the head-wearable apparatus **100** can receive input from a user of the head-wearable apparatus **100**.

(36) FIG. 1B illustrates the head-wearable apparatus **100** from the perspective of a user while wearing the head-wearable apparatus **100**. For clarity, a number of the elements shown in FIG. 1A have been omitted. As described in FIG. 1A, the head-wearable apparatus **100** shown in FIG. 1B includes left optical element **140** and right optical element **144** secured within the left optical element holder **132** and the right optical element holder **136** respectively.

(37) The head-wearable apparatus **100** includes right forward optical assembly **130** comprising a left near eye display **150**, a right near eye display **134**, and a left forward optical assembly **142** including a left projector **146** and a right projector **152**.

(38) In some examples, the near eye displays are waveguides. The waveguides include reflective or

diffractive structures (e.g., gratings and/or optical elements such as mirrors, lenses, or prisms). Light **138** emitted by the right projector **152** encounters the diffractive structures of the waveguide of the right near eye display **134**, which directs the light towards the right eye of a user to provide an image on or in the right optical element **144** that overlays the view of the real-world scene seen by the user. Similarly, light **148** emitted by the left projector **146** encounters the diffractive structures of the waveguide of the left near eye display **150**, which directs the light towards the left eye of a user to provide an image on or in the left optical element **140** that overlays the view of the real-world scene seen by the user. The combination of a Graphical Processing Unit, an image display driver, the right forward optical assembly **130**, the left forward optical assembly **142**, left optical element **140**, and the right optical element **144** provide an optical engine of the head-wearable apparatus **100**. The head-wearable apparatus **100** uses the optical engine to generate an overlay of the real-world scene view of the user including display of a user interface to the user of the head-wearable apparatus **100**.

(39) It will be appreciated however that other display technologies or configurations may be utilized within an optical engine to display an image to a user in the user's field of view. For example, instead of a projector and a waveguide, an LCD, LED or other display panel or surface may be provided.

(40) In use, a user of the head-wearable apparatus **100** will be presented with information, content and various user interfaces on the near eye displays. As described in more detail herein, the user can then interact with the head-wearable apparatus **100** using a touchpad **126** and/or the button **128**, voice inputs or touch inputs on an associated device (e.g. mobile device **414** illustrated in FIG. 4), and/or hand movements, locations, and positions recognized by the head-wearable apparatus **100**.

(41) In some examples, an optical engine of an XR system is incorporated into a lens that is contact with a user's eye, such as a contact lens or the like. The XR system generates images of an XR experience using the contact lens.

(42) In some examples, the head-wearable apparatus **100** comprises an XR system. In some examples, the head-wearable apparatus **100** is a component of an XR system including additional computational components. In some examples, the head-wearable apparatus **100** is a component in an XR system comprising additional user input systems or devices.

Machine Architecture

(43) FIG. 2 is a diagrammatic representation of the machine **200** within which instructions **202** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **200** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **202** may cause the machine **200** to execute any one or more of the methods described herein. The instructions **202** transform the general, non-programmed machine **200** into a particular machine **200** programmed to carry out the described and illustrated functions in the manner described. The machine **200** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **200** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **200** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **202**, sequentially or otherwise, that specify actions to be taken by the machine **200**. Further, while a single machine **200** is illustrated, the term "machine" shall also be taken to include a collection of machines that individually or jointly execute the instructions **202** to perform any one or more of the methodologies discussed herein. The machine **200**, for example, may comprise the computing system **502** or any one of

multiple server devices forming part of the interaction server system **510**. In some examples, the machine **200** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

(44) The machine **200** may include processors **204**, memory **206**, and input/output I/O components **208**, which may be configured to communicate with each other via a bus **210**. In an example, the processors **204** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **212** and a processor **214** that execute the instructions **202**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. 2 shows multiple processors **204**, the machine **200** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

(45) The memory **206** includes a main memory **216**, a static memory **240**, and a storage unit **218**, both accessible to the processors **204** via the bus **210**. The main memory **206**, the static memory **240**, and storage unit **218** store the instructions **202** embodying any one or more of the methodologies or functions described herein. The instructions **202** may also reside, completely or partially, within the main memory **216**, within the static memory **240**, within machine-readable medium **220** within the storage unit **218**, within at least one of the processors **204** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **200**.

(46) The I/O components **208** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **208** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **208** may include many other components that are not shown in FIG. 2. In various examples, the I/O components **208** may include user output components **222** and user input components **224**. The user output components **222** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **224** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(47) In further examples, the I/O components **208** may include biometric components **226**, motion components **228**, environmental components **230**, or position components **232**, among a wide array of other components. For example, the biometric components **226** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion

components **228** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

(48) The environmental components **230** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), depth or distance sensors (e.g., sensors to determine a distance to an object or a depth in a 3D coordinate system of features of an object), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

(49) The position components **232** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(50) Communication may be implemented using a wide variety of technologies. The I/O components **208** further include communication components **234** operable to couple the machine **200** to a network **236** or devices **238** via respective coupling or connections. For example, the communication components **234** may include a network interface component or another suitable device to interface with the network **236**. In further examples, the communication components **234** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **238** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

(51) Moreover, the communication components **234** may detect identifiers or include components operable to detect identifiers. For example, the communication components **234** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **234**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

(52) The various memories (e.g., main memory **216**, static memory **240**, and memory of the processors **204**) and storage unit **218** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **202**), when executed by processors **204**, cause various operations to implement the disclosed examples.

(53) The instructions **202** may be transmitted or received over the network **236**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **234**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **202** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **238**.

(54) FIG. 3A illustrates a collaboration diagram of components of an XR system using hand-tracking for user input, FIG. 3B illustrates a process flow diagram of a method of using a pinch

gesture as user input, and FIG. 3C illustrates using a pinch gesture to rotate a virtual object, in accordance with some examples.

(55) Although a method of rotating virtual objects **300** of FIG. 3B depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel, in a different sequence, or by different components of an XR system, that does not materially affect the function of the method.

(56) The method of rotating virtual objects **300** is used by an XR system **317**, such as head-wearable apparatus **100** (of FIG. 1A), to provide a continuous real-time input modality to a user **314** of the XR system **317** where the user **314** interacts with an XR user interface **324** using hand gestures or hand poses being made by the user **314** using one or more of their hands **327**. The XR user interface **324** may be for a useful application such as a maintenance guide, an interactive map, an interactive tour guide, a tutorial, or the like. The application may also be an entertainment application such as a video game, an interactive video, or the like.

(57) In some examples, the XR system **317** detects pinch gestures being made by the user **314**. A pinch gesture is a hand movement made by a user using a thumb and index finger. It involves bringing the thumb and index finger close together, with the fingertips touching or nearly touching, while keeping the other fingers extended or slightly bent. When performing a pinch gesture, the thumb and index finger act as opposing forces, creating a pinching motion. This gesture resembles the action of grasping or picking up something small between the fingertips, simulating the motion of pinching an object. The pinch gesture is associated with physically manipulating objects or performing delicate tasks that require precision, like picking up a small item, adjusting the settings on a device, or using fine motor skills. In some examples, the pinch gesture may be made using digits other than the thumb and index finger such as the middle, ring, or little finger.

(58) In operation **301**, the XR system **317** generates the XR user interface **324** provided to the user **314**. For example, a user interface engine **313** (of FIG. 3A) includes XR user interface control logic **330** comprising a dialog script or the like that specifies a user interface dialog implemented by the XR user interface **324**. The XR user interface control logic **330** also comprises one or more actions that are to be taken by the XR system **317** based on detecting various dialog events such as user inputs input by the user **314**. The user interface engine **313** further includes an XR user interface object model **329**. The XR user interface object model **329** includes 3D coordinate data of one or more virtual objects, such as a virtual object **333** (of FIG. 3C), and 3D coordinate data of one or more pinch box colliders associated with the virtual object **333**, such as a rotate collider **337** (of FIG. 3D). The rotate collider **337** is a virtual object that a user **314** interacts with in order to adjust a parameter of the virtual object **333** such as, but not limited to, a rotation of the virtual object **333**. The XR user interface object model **329** also includes 3D graphics data of the virtual object **333** and the rotate collider **337**. The 3D graphics data is used by an optical engine **323** to generate the XR user interface **324** for display to the user **314**.

(59) In some examples, a shape of the rotate collider **337** can comprise one or more irregular or regular 3D geometric shapes such as, but not limited to, a regular or irregular polyhedron, a spheroid, a conoid shape, or the like. In some examples, a shape of the rotate collider **337** is generated using one or more 3D geometric features of the virtual object **333**.

(60) The user interface engine **313** generates XR user interface graphics data **318** using the XR user interface object model **329**. The XR user interface graphics data **318** includes image data of the one or more virtual objects **333** of the XR user interface **324**. The user interface engine **313** communicates the XR user interface graphics data **318** to an image display driver **319** of an optical engine **323** of the XR system **317**. The image display driver **319** receives the XR user interface graphics data **318** and generates display control signals **320** using the XR user interface graphics data **318**. The image display driver **319** uses the display control signals **320** to control the

operations of one or more optical assemblies **311** of the optical engine **323**. In response to the display control signals **320**, the one or more optical assemblies **311** generate visible images of the XR user interface **324** that are provided to the user **314**.

(61) In operation **302**, the XR system **317** detects a pinch gesture made by the user **314**. For example, the XR system **317** uses one or more hand-tracking cameras **325** to capture tracking image data **326** of gestures **321** being made by the user **314** using one or more of the user's hands **327**. The hand-tracking cameras **325** communicate the tracking image data **326** to a hand-tracking component **312** of a hand-tracking pipeline **322** of the XR system **317**.

(62) The hand-tracking component **312** receives the tracking image data **326** and generates hand-tracking data **316** using the tracking image data **326**. The hand-tracking data **316** comprises skeletal model data of one or more skeletal models of the one or more hands **327** of the user in a 3D coordinate system using landmark features extracted from the tracking image data **326**, and hand gesture categorization data of a gesture **321** being made by the user's one or more hands **327**. The skeletal models comprise skeletal model features that correspond to recognized visual landmarks of portions of the one or more hands **327** of the user **314**. In some examples, the hand-tracking data **316** includes landmark data such as landmark identification, a physical location of the landmark, links between joints of the user's fingers and categorization information of one or more landmarks associated with the one or more hands **327** of the user **314**. In some examples, the hand gesture categorization data includes an indication of a pinch gesture being made by one or more of the hands **327** of the user **314**.

(63) In some examples, the hand-tracking component **312** recognizes landmark features on portions of the one or more hands **327** of the user **314** captured in the tracking image data **326**. The hand-tracking component **312** extracts landmarks of the one or more hands **327** of the user **314** from the tracking image data **326** using computer vision methodologies including, but not limited to, Harris corner detection, Shi-Tomasi corner detection, Scale-Invariant Feature Transform (SIFT), Speeded-Up Robust Features (SURF), Features from Accelerated Segment Test (FAST), Oriented FAST and Rotated BRIEF (ORB), and the like.

(64) In some examples, the hand-tracking component **312** generates the hand gesture categorization data and the sequence of skeletal models of hand-tracking data **316** using the landmarks extracted from the tracking image data **326** using artificial intelligence methodologies and an ML hand-tracking model **315** that was previously generated using machine learning methodologies. In some examples, an ML hand-tracking model **315** comprises, but is not limited to, a neural network, a learning vector quantization network, a logistic regression model, a support vector machine, a random decision forest, a naive Bayes model, a linear discriminant analysis model, and a K-nearest neighbor model. In some examples, machine learning methodologies used to generate the ML hand-tracking model **315** may include, but are not limited to, supervised learning, unsupervised learning, semi-supervised learning, reinforcement learning, dimensionality reduction, self-learning, feature learning, sparse dictionary learning, and anomaly detection.

(65) In some examples, the hand-tracking component **312** generates the hand gesture categorization data and the sequence of skeletal models of the hand-tracking data **316** using the landmarks extracted from the tracking image data **326** using geometric methodologies.

(66) In some examples, a pinch gesture comprises a user touching the fingertip of the forefinger of one of their hands to the fingertip of the thumb of the same hand.

(67) In some examples, a pinch gesture is determined by detecting a sequence of proto hand poses or proto gestures as the user moves the fingertip of the forefinger of one of their hands closer to the fingertip of the thumb of the same hand.

(68) The hand-tracking component **312** of the XR system **317** communicates the hand-tracking data **316** to the user interface engine **313**. The user interface engine **313** detects that the user **314** is making a pinch gesture using the hand gesture categorization data of the hand-tracking data **316**.

(69) In some examples, other selection hand gestures or selection hand poses are used by a user to

indicate that the user's intent is to rotate virtual object. For example, selection hand poses or selection hand gestures may include, but are not limited to, a tip of one or more of digits held in apparent contact with an edge or surface of a virtual object, grasping a virtual object with two or more digits, and the like.

(70) FIG. 3D illustrates a 3D coordinate system, a virtual object, and a rotate collider, in accordance with some examples. In response to detecting that the user **314** is making a pinch gesture, the user interface engine **313** uses a collision detector **328** to determine whether the pinch gesture is being made within a rotate collider **337** of the virtual object **333**. The determination is made by the collision detector **328** using the skeletal model data of the hand-tracking data **316** and the 3D coordinate data of the rotate collider **337** included in the XR user interface object model **329**. For example, the user interface engine **313** determines the 3D coordinate data of a forefinger tip node and a thumb tip node of the skeletal model and generates a selection location collider using the 3D coordinate data of the forefinger tip node and the thumb tip node. The collision detector **328** determines whether the 3D geometries of the rotate collider **337** and the selection location collider collide by detecting an intersection between the 3D geometries of the rotate collider **337** and the selection location collider. In a case of a point collider, a collision is determined if the point collider is located within a volume of another collider.

(71) In some examples, a shape of the selection location collider comprises one or more irregular or regular 3D geometric shapes such as, but not limited to, a regular or irregular polyhedron, a spheroid, a conoid shape, or the like. In some examples, a shape of the selection location collider is generated using one or more features of the user's hand **327**.

(72) Referring to FIG. 3B and FIG. 3D, in operation **303**, the user interface engine **313** determines a set of 3D coordinates of a first selection location **339** of the pinch gesture as the user **314** is making the pinch gesture. The 3D coordinates are in reference to a 3D coordinate system of the XR user interface **324** as represented by X axis **335**, Y axis **334**, and Z axis **336**. In some examples, the 3D coordinates of the first selection location **339** are determined using a center point of the selection location collider. For example, XR system determines the 3D coordinate data of the first selection location **339** using the 3D coordinate data of the center point of the selection location collider. In some examples, the 3D coordinates of the first selection location **339** are determined using the 3D coordinate data of an edge, vertex, or surface of the selection location collider. In some examples, the 3D coordinate data of the first selection location **339** are determined using an intersection of the selection location collider and the rotate collider **337**. In some examples, the user interface engine **313** uses a center point of the rotate collider **337** to determine the first selection location **339** of the pinch gesture. A location and orientation of the virtual object **333** is determined by an orientation vector **340** of the virtual object **333**. The orientation vector **340** may be stored as a quaternion in the 3D coordinate system.

(73) FIG. 3E illustrates a view of the virtual object **333** in a 3D coordinate system having as its origin a virtual object center point **338** of the virtual object **333**, in accordance with some examples. An X' axis **347**, Y' axis **346**, and Z' axis **348** extend through the virtual object center point **338**. The Z' axis **348** is parallel to a Z component of the orientation vector **340**. The virtual object **333** is to be rotated about the Y' axis **346**.

(74) FIG. 3F illustrates a projection of the virtual object **333** onto a 2D plane defined by the Z' axis **348** and the X' axis **347**, in accordance with some examples. The virtual object **333** is viewed as if looking down along the Y' axis **346** (of FIG. 3E). A plane defined by the X' axis **347** and the Z' axis **348** intersects the virtual object **333** and the virtual object center point **338**. The rotate collider **337** and the first selection location **339** are also shown as being projected onto the plane defined by the Z' axis **348** and the X' axis **347**.

(75) Referring to FIG. 3B and FIG. 3E, in operation **304**, the user interface engine **313** determines an offset angle **341** using the 3D coordinate data of the virtual object center point **338** of the virtual object **333**, the orientation vector **340**, and the 3D coordinate data of the first selection location

339. For example, the user interface engine 313 projects a first selection location vector 342 from the virtual object center point 338 of the virtual object 333 to the first selection location 339 of the pinch gesture and determines the offset angle 341 using the first selection location vector 342 and the orientation vector 340.

(76) FIG. 3G illustrates a translation 350 of a selection location, in accordance with some examples. Referring to FIG. 3G and FIG. 3B, in operation 305, the user interface engine 313 detects a translation 350 of the first selection location 339 of the pinch gesture being made by the user 314 to a second selection location 345. For example, the user interface engine 313 continuously receives the hand-tracking data 316 from the hand-tracking pipeline 322 and determines if the user 314 is still holding the pinch gesture and if the user 314 has moved their hand 327 to a second selection location 345, thus translating the first selection location 339 to the second selection location 345.

(77) In operation 306, in response to detecting that the user 314 has moved a pinch gesture to the second selection location 345, the user interface engine 313 determines a rotation angle 344 using the first selection location 339 of the pinch gesture and the second selection location 345 of the pinch gesture. For example, the user interface engine 313 determines a second selection location vector 343 using the virtual object center point 338 and the second selection location 345. The user interface engine 313 uses the first selection location vector 342 and the second selection location vector 343 to determine the rotation angle 344.

(78) FIG. 3H illustrates a rotated virtual object projected onto a 2D plane, and FIG. 3I illustrates a rotated virtual object in an XR user interface, in accordance with some examples. Referring to FIG. 3H, FIG. 3I, and FIG. 3B, in operation 307, the user interface engine 313 rotates the virtual object 333 using the rotation angle 344. For example, the user interface engine 313 generates a rotation matrix using the rotation angle 344 and rotates the orientation vector 340 around the Y' axis 346 (of FIG. 3E) using the rotation matrix. This maintains the offset angle 341 between the orientation vector 340 and the current selection location, namely second selection location 345.

(79) In operation 308, the user interface engine 313 uses the optical engine 323 to redisplay the virtual object 333 using the rotated orientation vector 340 of the of the virtual object 333 in the 324.

(80) In operation 309, the user interface engine 313 determines whether the pinch gesture is still being held by the user 314. For example, the XR system 317 captures additional tracking image data 326 of the hand 327 of the user 314. The XR system generates additional hand-tracking data 316 using the additional tracking image data 326 as described herein. Included in the additional hand-tracking data 316 is additional hand gesture or hand gesture categorization data. The user interface engine 313 detects whether the pinch gesture is still being held by the user using the additional hand gesture or hand gesture categorization data of the hand-tracking data 316.

(81) In some examples, a release of a pinch gesture comprises a user moving the fingertip of the forefinger of one of their hands away from the fingertip of the thumb of the same hand.

(82) In some examples, a release of the pinch gesture is determined by detecting a sequence of proto hand poses or proto gestures as the user moves the fingertip of the forefinger of one of their hands away from the fingertip of the thumb of the same hand.

(83) In response to determining that the pinch gesture is still being held, the XR system transitions 331 to operation 303 and repeats the process of the rotating the virtual object 333 as described herein. In response to determining that the pinch gesture is not being held, the XR system transitions 332 to operation 310 and ends.

(84) Referring to FIG. 3E, in some examples, the user interface engine 313 rotates the virtual object 333 around the X' axis 347. For example, the user interface engine 313 projects the first selection location 339 and the virtual object 333 onto a plane defined by the Y' axis 346 and the Z' axis 348. The user interface engine 313 determines a translation of the first selection location 339 to a second selection location on the plane defined by the Y' axis 346 and the Z' axis 348 and uses the first

selection location **339** and the second selection location to rotate the virtual object **333** around the X' axis **347**.

(85) In some examples, the user interface engine **313** rotates the virtual object **333** around the Z' axis **348**. For example, the user interface engine **313** projects the first selection location **339** and the virtual object **333** onto a plane defined by the Y' axis **346** and the X' axis **347**. The user interface engine **313** determines a translation of the first selection location **339** to a second selection location on the plane defined by the Y' axis **346** and the X' axis **347** and uses the first selection location **339** and the second selection location to rotate the virtual object **333** around the Z' axis **348**.

(86) In some examples, the user interface engine **313** determines an axis around which to rotate the virtual object **333** using a direction of a translation of the first selection location **339** by the user **314**. For example, the user interface engine **313** detects a second selection location and determines which axis, the X' axis **347**, the Y' axis **346**, or the Z' axis **348**, showed a maximum difference as between the three axis. If the maximum change in the second selection location was along the X' axis **347**, the user interface engine **313** determines to rotate the virtual object **333** around the Y' axis **346**. If the maximum change in the second selection location was along the Y' axis **346**, the user interface engine **313** determines to rotate the virtual object **333** around the Z' axis **348**. If the maximum change in the second selection location was along the Z' axis **348**, the user interface engine **313** determines to rotate the virtual object **333** around the X' axis **347**.

(87) In some examples, the user interface engine **313** provides a plurality of rotate colliders **349** corresponding respectively to the X' axis **347**, the Y' axis **346**, and the Z' axis **348**. The user interface engine **313** determines which axis to rotate the part around using a user selection of a rotate collider of the plurality of rotate colliders.

(88) In some examples, the XR system performs the functions of the hand-tracking pipeline **322**, the user interface engine **313**, and the optical engine **323** utilizing various APIs and system libraries.

System with Head-Wearable Apparatus

(89) FIG. **4** illustrates a system **400** including a head-wearable apparatus **100** with a selector input device, according to some examples. FIG. **4** is a high-level functional block diagram of an example head-wearable apparatus **100** communicatively coupled to a mobile device **414** and various server systems **404** (e.g., the interaction server system **510**) via various networks **508**.

(90) The head-wearable apparatus **100** includes one or more cameras, each of which may be, for example, one or more camera **408**, a light emitter **410**, and one or more wide-spectrum cameras **412**.

(91) The mobile device **414** connects with head-wearable apparatus **100** using both a low-power wireless connection **416** and a high-speed wireless connection **418**. The mobile device **414** is also connected to the server system **404** and the network **406**.

(92) The head-wearable apparatus **100** further includes two image displays of the image display of optical assembly **420**. The two image displays of optical assembly **420** include one associated with the left lateral side and one associated with the right lateral side of the head-wearable apparatus **100**. The head-wearable apparatus **100** also includes an image display driver **422**, and a GPU **424**. The image display of optical assembly **420**, image display driver **422**, and GPU **424** constitute an optical engine of the head-wearable apparatus **100**. The image display of optical assembly **420** is for presenting images and videos, including an image that can include a graphical user interface to a user of the head-wearable apparatus **100**.

(93) The image display driver **422** commands and controls the image display of optical assembly **420**. The image display driver **422** may deliver image data directly to the image display of optical assembly **420** for presentation or may convert the image data into a signal or data format suitable for delivery to the image display device. For example, the image data may be video data formatted according to compression formats, such as H.264 (MPEG-4 Part 10), HEVC, Theora, Dirac, RealVideo RV40, VP8, VP9, or the like, and still image data may be formatted according to

compression formats such as Portable Network Group (PNG), Joint Photographic Experts Group (JPEG), Tagged Image File Format (TIFF) or exchangeable image file format (EXIF) or the like. (94) The head-wearable apparatus **100** includes a frame and stems (or temples) extending from a lateral side of the frame. The head-wearable apparatus **100** further includes a user input device **430** (e.g., touch sensor or push button), including an input surface on the head-wearable apparatus **100**. The user input device **430** (e.g., touch sensor or push button) is to receive from the user an input selection to manipulate the graphical user interface of the presented image.

(95) The components shown in FIG. 4 for the head-wearable apparatus **100** are located on one or more circuit boards, for example a PCB or flexible PCB, in the rims or temples. Alternatively, or additionally, the depicted components can be located in the chunks, frames, hinges, or bridge of the head-wearable apparatus **100**. Left and right cameras **408** can include digital camera elements such as a complementary metal oxide—semiconductor (CMOS) image sensor, charge-coupled device, camera lenses, or any other respective visible or light-capturing elements that may be used to capture data, including images of scenes with unknown objects.

(96) The head-wearable apparatus **100** includes a memory **402**, which stores instructions to perform a subset or all of the functions described herein. The memory **402** can also include storage device.

(97) As shown in FIG. 4, the high-speed circuitry **428** includes a high-speed processor **432**, a memory **402**, and high-speed wireless circuitry **434**. In some examples, the image display driver **422** is coupled to the high-speed circuitry **428** and operated by the high-speed processor **432** in order to drive the left and right image displays of the image display of optical assembly **420**. The high-speed processor **432** may be any processor capable of managing high-speed communications and operation of any general computing system needed for the head-wearable apparatus **100**. The high-speed processor **432** includes processing resources needed for managing high-speed data transfers on a high-speed wireless connection **418** to a wireless local area network (WLAN) using the high-speed wireless circuitry **434**. In certain examples, the high-speed processor **432** executes an operating system such as a LINUX operating system or other such operating system of the head-wearable apparatus **100**, and the operating system is stored in the memory **402** for execution. In addition to any other responsibilities, the high-speed processor **432** executing a software architecture for the head-wearable apparatus **100** is used to manage data transfers with high-speed wireless circuitry **434**. In certain examples, the high-speed wireless circuitry **434** is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to herein as WiFi. In some examples, other high-speed communications standards may be implemented by the high-speed wireless circuitry **434**.

(98) The low-power wireless circuitry **436** and the high-speed wireless circuitry **434** of the head-wearable apparatus **100** can include short-range transceivers (Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi). Mobile device **414**, including the transceivers communicating via the low-power wireless connection **416** and the high-speed wireless connection **418**, may be implemented using details of the architecture of the head-wearable apparatus **100**, as can other elements of the network **406**.

(99) The memory **402** includes any storage device capable of storing various data and applications, including, among other things, camera data generated by the left and right cameras **408**, the wide-spectrum cameras **412**, and the GPU **424**, as well as images generated for display by the image display driver **422** on the image displays of the image display of optical assembly **420**. While the memory **402** is shown as integrated with high-speed circuitry **428**, in some examples, the memory **402** may be an independent standalone element of the head-wearable apparatus **100**. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor **432** from the GPU **424** or the low-power processor **438** to the memory **402**. In some examples, the high-speed processor **432** may manage addressing of the memory **402** such that the low-power processor **438** will boot the high-speed processor **432** any time that a read or write operation involving memory **402** is needed.

(100) As shown in FIG. 4, the low-power processor **438** or high-speed processor **432** of the head-wearable apparatus **100** can be coupled to the camera (camera **408**, light emitter **410**, or wide-spectrum cameras **412**), the image display driver **422**, the user input device **430** (e.g., touch sensor or push button), and the memory **402**.

(101) The head-wearable apparatus **100** is connected to a host computer. For example, the head-wearable apparatus **100** is paired with the mobile device **414** via the high-speed wireless connection **418** or connected to the server system **404** via the network **406**. The server system **404** may be one or more computing devices as part of a service or network computing system, for example, that includes a processor, a memory, and network communication interface to communicate over the network **406** with the mobile device **414** and the head-wearable apparatus **100**.

(102) The mobile device **414** includes a processor and a network communication interface coupled to the processor. The network communication interface allows for communication over the network **406**, low-power wireless connection **416**, or high-speed wireless connection **418**. Mobile device **414** can further store at least portions of the instructions for generating binaural audio content in the mobile device **414**'s memory to implement the functionality described herein.

(103) Output components of the head-wearable apparatus **100** include visual components, such as a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light-emitting diode (LED) display, a projector, or a waveguide. The image displays of the optical assembly are driven by the image display driver **422**. The output components of the head-wearable apparatus **100** further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the head-wearable apparatus **100**, the mobile device **414**, and server system **404**, such as the user input device **430**, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(104) The head-wearable apparatus **100** may also include additional peripheral device elements. Such peripheral device elements may include biometric sensors, additional sensors, or display elements integrated with the head-wearable apparatus **100**. For example, peripheral device elements may include any I/O components including output components, motion components, position components, or any other such elements described herein.

(105) For example, the biometric components include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), Wi-Fi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over low-power wireless connections **416** and high-speed wireless connection **418** from the mobile device **414** via the low-power wireless circuitry **436** or high-speed wireless circuitry **434**.

Networked Computing Environment

(106) FIG. 5 is a block diagram showing an example interaction system **500** for facilitating

interactions (e.g., exchanging text messages, conducting text audio and video calls, or playing games) over a network. The interaction system **500** includes multiple XR systems **502**, each of which hosts multiple applications, including an interaction client **504** and other applications **506**. Each interaction client **504** is communicatively coupled, via one or more communication networks including a network **508** (e.g., the Internet), to other instances of the interaction client **504** (e.g., hosted on respective other XR systems **502**), an interaction server system **510** and third-party servers **512**). An interaction client **504** can also communicate with locally hosted applications **506** using Applications Program Interfaces (APIs).

(107) Each computing system **502** may comprise one or more user devices, such as a mobile device **414**, head-wearable apparatus **100**, and a computer client device **514** that are communicatively connected to exchange data and messages.

(108) An interaction client **504** interacts with other interaction clients **504** and with the interaction server system **510** via the network **508**. The data exchanged between the interaction clients **504** (e.g., interactions **516**) and between the interaction clients **504** and the interaction server system **510** includes functions (e.g., commands to invoke functions) and payload data (e.g., text, audio, video, or other multimedia data).

(109) The interaction server system **510** provides server-side functionality via the network **508** to the interaction clients **504**. While certain functions of the interaction system **500** are described herein as being performed by either an interaction client **504** or by the interaction server system **510**, the location of certain functionality either within the interaction client **504** or the interaction server system **510** may be a design choice. For example, it may be technically preferable to initially deploy particular technology and functionality within the interaction server system **510** but to later migrate this technology and functionality to the interaction client **504** where a computing system **502** has sufficient processing capacity.

(110) The interaction server system **510** supports various services and operations that are provided to the interaction clients **504**. Such operations include transmitting data to, receiving data from, and processing data generated by the interaction clients **504**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information. Data exchanges within the interaction system **500** are invoked and controlled through functions available via user interfaces (UIs) of the interaction clients **504**.

(111) Turning now specifically to the interaction server system **510**, an Application Program Interface (API) server **518** is coupled to and provides programmatic interfaces to Interaction servers **520**, making the functions of the Interaction servers **520** accessible to interaction clients **504**, other applications **506** and third-party server **512**. The interaction servers **520** are communicatively coupled to a database server **522**, facilitating access to a database **524** that stores data associated with interactions processed by the Interaction servers **520**. Similarly, a web server **526** is coupled to the interaction servers **520** and provides web-based interfaces to the Interaction servers **520**. To this end, the web server **526** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

(112) The Application Program Interface (API) server **518** receives and transmits interaction data (e.g., commands and message payloads) between the interaction servers **520** and the XR systems **502** (and, for example, interaction clients **504** and other application **506**) and the third-party server **512**. Specifically, the Application Program Interface (API) server **518** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the interaction client **504** and other applications **506** to invoke functionality of the Interaction servers **520**. The Application Program Interface (API) server **518** exposes various functions supported by the Interaction servers **520**, including account registration; login functionality; the sending of interaction data, via the Interaction servers **520**, from a particular interaction client **504** to another interaction client **504**; the communication of media files (e.g., images or video) from an interaction client **504** to the

interaction servers **520**; the settings of a collection of media data (e.g., a story); the retrieval of a list of friends of a user of a computing system **502**; the retrieval of messages and content; the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph); the location of friends within a social graph; and opening an application event (e.g., relating to the interaction client **504**).

(113) The interaction servers **520** host multiple systems and subsystems, described below with reference to FIG. 7.

(114) Returning to the interaction client **504**, features and functions of an external resource (e.g., a linked application **506** or applet) are made available to a user via an interface of the interaction client **504**. In this context, “external” refers to the fact that the application **506** or applet is external to the interaction client **504**. The external resource is often provided by a third party but may also be provided by the creator or provider of the interaction client **504**. The interaction client **504** receives a user selection of an option to launch or access features of such an external resource. The external resource may be the application **506** installed on the computing system **502** (e.g., a “native app”), or a small-scale version of the application (e.g., an “applet”) that is hosted on the computing system **502** or remote of the computing system **502** (e.g., on third-party servers **512**). The small-scale version of the application includes a subset of features and functions of the application (e.g., the full-scale, native version of the application) and is implemented using a markup-language document. In some examples, the small-scale version of the application (e.g., an “applet”) is a web-based, markup-language version of the application and is embedded in the interaction client **504**. In addition to using markup-language documents (e.g., a *.ml file), an applet may incorporate a scripting language (e.g., a *.js file or a .json file) and a style sheet (e.g., a *.ss file).

(115) In response to receiving a user selection of the option to launch or access features of the external resource, the interaction client **504** determines whether the selected external resource is a web-based external resource or a locally-installed application **506**. In some cases, applications **506** that are locally installed on the computing system **502** can be launched independently of and separately from the interaction client **504**, such as by selecting an icon corresponding to the application **506** on a home screen of the computing system **502**. Small-scale versions of such applications can be launched or accessed via the interaction client **504** and, in some examples, no or limited portions of the small-scale application can be accessed outside of the interaction client **504**. The small-scale application can be launched by the interaction client **504** receiving, from a third-party server **512** for example, a markup-language document associated with the small-scale application and processing such a document.

(116) In response to determining that the external resource is a locally-installed application **506**, the interaction client **504** instructs the computing system **502** to launch the external resource by executing locally-stored code corresponding to the external resource. In response to determining that the external resource is a web-based resource, the interaction client **504** communicates with the third-party servers **512** (for example) to obtain a markup-language document corresponding to the selected external resource. The interaction client **504** then processes the obtained markup-language document to present the web-based external resource within a user interface of the interaction client **504**.

(117) The interaction client **504** can notify a user of the computing system **502**, or other users related to such a user (e.g., “friends”), of activity taking place in one or more external resources. For example, the interaction client **504** can provide participants in a conversation (e.g., a chat session) in the interaction client **504** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using respective interaction clients **504**, with the ability to share an item, status, state, or location in an external resource in a chat session with one or more members of a group of users. The shared

item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the interaction client **504**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

(118) The interaction client **504** can present a list of the available external resources (e.g., applications **506** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the application **506** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

Data Architecture

(119) FIG. **6** is a schematic diagram illustrating data structures **600**, which may be stored in the database **604** of the interaction server system **510**, according to certain examples. While the content of the database **604** is shown to comprise multiple tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

(120) The database **604** includes message data stored within a message table **606**. This message data includes, for any particular message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **606**, are described below with reference to FIG. **6**.

(121) An entity table **608** stores entity data, and is linked (e.g., referentially) to an entity graph **610** and profile data **602**. Entities for which records are maintained within the entity table **608** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the interaction server system **510** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

(122) The entity graph **610** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization), interest-based, or activity-based, merely for example. Certain relationships between entities may be unidirectional, such as a subscription by an individual user to digital content of a commercial or publishing user (e.g., a newspaper or other digital media outlet, or a brand). Other relationships may be bidirectional, such as a “friend” relationship between individual users of the interaction system **500**.

(123) Certain permissions and relationships may be attached to each relationship, and also to each direction of a relationship. For example, a bidirectional relationship (e.g., a friend relationship between individual users) may include authorization for the publication of digital content items between the individual users, but may impose certain restrictions or filters on the publication of such digital content items (e.g., based on content characteristics, location data or time of day data). Similarly, a subscription relationship between an individual user and a commercial user may impose different degrees of restrictions on the publication of digital content from the commercial user to the individual user, and may significantly restrict or block the publication of digital content from the individual user to the commercial user. A particular user, as an example of an entity, may record certain restrictions (e.g., by way of privacy settings) in a record for that entity within the entity table **608**. Such privacy settings may be applied to all types of relationships within the context of the interaction system **500**, or may selectively be applied to only certain types of relationships.

(124) The profile data **602** stores multiple types of profile data about a particular entity. The profile data **602** may be selectively used and presented to other users of the interaction system **500** based on privacy settings specified by a particular entity. Where the entity is an individual, the profile

data **602** includes, for example, a username, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the interaction system **500**, and on map interfaces displayed by interaction clients **504** to other users. The collection of avatar representations may include “status avatars,” which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

(125) Where the entity is a group, the profile data **602** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

(126) The database **604** also stores augmentation data, such as overlays or filters, in an augmentation table **612**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **614**) and images (for which data is stored in an image table **616**).

(127) Filters, in some examples, are overlays that are displayed as overlaid on an image or video during presentation to a message receiver. Filters may be of various types, including user-selected filters from a set of filters presented to a message sender by the interaction client **504** when the message sender is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a message sender based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the interaction client **504**, based on geolocation information determined by a Global Positioning System (GPS) unit of the computing system **502**.

(128) Another type of filter is a data filter, which may be selectively presented to a message sender by the interaction client **504** based on other inputs or information gathered by the computing system **502** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a message sender is traveling, battery life for a computing system **502**, or the current time.

(129) Other augmentation data that may be stored within the image table **616** includes augmented reality content items (e.g., corresponding to applying Lenses or augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

(130) As described above, augmentation data includes AR, VR, and mixed reality (MR) content items, overlays, image transformations, images, and modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of the computing system **502** and then displayed on a screen of the computing system **502** with the modifications. This also includes modifications to stored content, such as video clips in a collection or group that may be modified. For example, in a computing system **502** with access to multiple augmented reality content items, a user can use a single video clip with multiple augmented reality content items to see how the different augmented reality content items will modify the stored clip. Similarly, real-time video capture may use modifications to show how video images currently being captured by sensors of a computing system **502** would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different augmented reality content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

(131) Data and various systems using augmented reality content items or other such transform systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as

they are tracked. In various examples, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects, and using transformations and animated textures of the model within the video to achieve the transformation. In some examples, tracking of points on an object may be used to place an image or texture (which may be two-dimensional or three-dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). Augmented reality content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

(132) Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device, or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

(133) In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of an object's elements, characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh.

(134) In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification, properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing the color of areas; removing some part of areas from the frames of the video stream; including new objects into areas that are based on a request for modification; and modifying or distorting the elements of an area or object. In various examples, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

(135) In some examples of a computer animation model to transform image data using face detection, the face is detected on an image using a specific face detection algorithm (e.g., Viola-Jones). Then, an Active Shape Model (ASM) algorithm is applied to the face region of an image to detect facial feature reference points.

(136) Other methods and algorithms suitable for face detection can be used. For example, in some examples, visual features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an

eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

(137) A transformation system can capture an image or video stream on a client device (e.g., the computing system **502**) and perform complex image manipulations locally on the computing system **502** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the computing system **502**.

(138) In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using the computing system **502** having a neural network operating as part of an interaction client **504** operating on the computing system **502**. The transformation system operating within the interaction client **504** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that are the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transform system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user interface displayed on the computing system **502** as soon as the image or video stream is captured and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured, and the selected modification icon remains toggled. Machine-taught neural networks may be used to enable such modifications.

(139) The graphical user interface, presenting the modification performed by the transform system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various examples, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browsing to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some examples, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

(140) A story table **618** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **608**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of

the interaction client **504** may include an icon that is user-selectable to enable a message sender to add specific content to his or her personal story.

(141) A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the interaction client **504**, to contribute content to a particular live story. The live story may be identified to the user by the interaction client **504**, based on his or her location. The end result is a “live story” told from a community perspective.

(142) A further type of content collection is known as a “location story,” which enables a user whose computing system **502** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end-user belongs to a specific organization or other entity (e.g., is a student on the university campus).

(143) As mentioned above, the video table **614** stores video data that, in some examples, is associated with messages for which records are maintained within the message table **606**. Similarly, the image table **616** stores image data associated with messages for which message data is stored in the entity table **608**. The entity table **608** may associate various augmentations from the augmentation table **612** with various images and videos stored in the image table **616** and the video table **614**.

(144) The databases **604** also includes social network information collected by the social network system **722**.

System Architecture

(145) FIG. 7 is a block diagram illustrating further details regarding the interaction system **500**, according to some examples. Specifically, the interaction system **500** is shown to comprise the interaction client **504** and the Interaction servers **520**. The interaction system **500** embodies multiple subsystems, which are supported on the client-side by the interaction client **504** and on the server-side by the Interaction servers **520**. Example subsystems are discussed below.

(146) An image processing system **702** provides various functions that enable a user to capture and augment (e.g., augment or otherwise modify or edit) media content associated with a message.

(147) A camera system **704** includes control software (e.g., in a camera application) that interacts with and controls hardware camera hardware (e.g., directly or via operating system controls) of the computing system **502** to modify and augment real-time images captured and displayed via the interaction client **504**.

(148) The augmentation system **706** provides functions related to the generation and publishing of augmentations (e.g., media overlays) for images captured in real-time by cameras of the computing system **502** or retrieved from memory of the computing system **502**. For example, the augmentation system **706** operatively selects, presents, and displays media overlays (e.g., an image filter or an image lens) to the interaction client **504** for the augmentation of real-time images received via the camera system **704** or stored images retrieved from memory **402** of a computing system **502**. These augmentations are selected by the augmentation system **706** and presented to a user of an interaction client **504**, based on a number of inputs and data, such as for example: Geolocation of the computing system **502**; and Social network information of the user of the computing system **502**.

(149) An augmentation may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo or video) at computing system **502** for communication in a message, or applied to video content, such as a video content stream or feed

transmitted from an interaction client **504**. As such, the image processing system **702** may interact with, and support, the various subsystems of the communication system **708**, such as the messaging system **710** and the video communication system **712**.

(150) A media overlay may include text or image data that can be overlaid on top of a photograph taken by the computing system **502** or a video stream produced by the computing system **502**. In some examples, the media overlay may be a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In further examples, the image processing system **702** uses the geolocation of the computing system **502** to identify a media overlay that includes the name of a merchant at the geolocation of the computing system **502**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the databases **524** and accessed through the database server **522**.

(151) The image processing system **702** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The image processing system **702** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

(152) The augmentation creation system **714** supports augmented reality developer platforms and includes an application for content creators (e.g., artists and developers) to create and publish augmentations (e.g., augmented reality experiences) of the interaction client **504**. The augmentation creation system **714** provides a library of built-in features and tools to content creators including, for example custom shaders, tracking technology, and templates.

(153) In some examples, the augmentation creation system **714** provides a merchant-based publication platform that enables merchants to select a particular augmentation associated with a geolocation via a bidding process. For example, the augmentation creation system **714** associates a media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.

(154) A communication system **708** is responsible for enabling and processing multiple forms of communication and interaction within the interaction system **500** and includes a messaging system **710**, an audio communication system **716**, and a video communication system **712**. The messaging system **710** is responsible for enforcing the temporary or time-limited access to content by the interaction clients **504**. The messaging system **710** incorporates multiple timers (e.g., within an ephemeral timer system **718**) that, based on duration and display parameters associated with a message or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the interaction client **504**. Further details regarding the operation of the ephemeral timer system **718** are provided below. The audio communication system **716** enables and supports audio communications (e.g., real-time audio chat) between multiple interaction clients **504**. Similarly, the video communication system **712** enables and supports video communications (e.g., real-time video chat) between multiple interaction clients **504**.

(155) A user management system **720** is operationally responsible for the management of user data and profiles, and includes a social network system **722** that maintains social network information regarding relationships between users of the interaction system **500**.

(156) A collection management system **724** is operationally responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **724** may also be responsible for publishing an icon that provides notification of a particular collection to the user interface of the interaction client **504**. The collection

management system **724** includes a curation function that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **724** employs machine vision (or image recognition technology) and content rules to curate a content collection automatically. In certain examples, compensation may be paid to a user to include user-generated content into a collection. In such cases, the collection management system **724** operates to automatically make payments to such users to use their content.

(157) A map system **726** provides various geographic location functions and supports the presentation of map-based media content and messages by the interaction client **504**. For example, the map system **726** enables the display of user icons or avatars (e.g., stored in profile data **602**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the interaction system **500** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the interaction client **504**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the interaction system **500** via the interaction client **504**, with this location and status information being similarly displayed within the context of a map interface of the interaction client **504** to selected users.

(158) A game system **728** provides various gaming functions within the context of the interaction client **504**. The interaction client **504** provides a game interface providing a list of available games that can be launched by a user within the context of the interaction client **504** and played with other users of the interaction system **500**. The interaction system **500** further enables a particular user to invite other users to participate in the play of a specific game by issuing invitations to such other users from the interaction client **504**. The interaction client **504** also supports audio, video, and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

(159) An external resource system **730** provides an interface for the interaction client **504** to communicate with remote servers (e.g., third-party servers **512**) to launch or access external resources, i.e., applications or applets. Each third-party server **512** hosts, for example, a markup language (e.g., HTML5) based application or a small-scale version of an application (e.g., game, utility, payment, or ride-sharing application). The interaction client **504** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the third-party servers **512** associated with the web-based resource. Applications hosted by third-party servers **512** are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the Interaction servers **520**. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. The Interaction servers **520** host a JavaScript library that provides a given external resource access to specific user data of the interaction client **504**. HTML5 is an example of technology for programming games, but applications and resources programmed based on other technologies can be used.

(160) To integrate the functions of the SDK into the web-based resource, the SDK is downloaded by the third-party server **512** from the Interaction servers **520** or is otherwise received by the third-party server **512**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the interaction client **504** into the web-based resource.

(161) The SDK stored on the interaction server system **510** effectively provides the bridge between an external resource (e.g., applications **506** or applets) and the interaction client **504**. This gives the user a seamless experience of communicating with other users on the interaction client **504** while

also preserving the look and feel of the interaction client **504**. To bridge communications between an external resource and an interaction client **504**, the SDK facilitates communication between third-party servers **512** and the interaction client **504**. A WebViewJavaScriptBridge running on a computing system **502** establishes two one-way communication channels between an external resource and the interaction client **504**. Messages are sent between the external resource and the interaction client **504** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

(162) By using the SDK, not all information from the interaction client **504** is shared with third-party servers **512**. The SDK limits which information is shared based on the needs of the external resource. Each third-party server **512** provides an HTML5 file corresponding to the web-based external resource to Interaction servers **520**. The Interaction servers **520** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the interaction client **504**. Once the user selects the visual representation or instructs the interaction client **504** through a GUI of the interaction client **504** to access features of the web-based external resource, the interaction client **504** obtains the HTML5 file and instantiates the resources to access the features of the web-based external resource.

(163) The interaction client **504** presents a graphical user interface (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the interaction client **504** determines whether the launched external resource has been previously authorized to access user data of the interaction client **504**. In response to determining that the launched external resource has been previously authorized to access user data of the interaction client **504**, the interaction client **504** presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the interaction client **504**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the interaction client **504** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the interaction client **504** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the interaction client **504**. The external resource is authorized by the interaction client **504** to access the user data under an OAuth 2 framework.

(164) The interaction client **504** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale applications (e.g., an application **506**) are provided with access to a first type of user data (e.g., two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of applications (e.g., web-based versions of applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

(165) An advertisement system **732** operationally enables the purchasing of advertisements by third parties for presentation to end-users via the interaction clients **504** and also handles the delivery and presentation of these advertisements.

Software Architecture

(166) FIG. **8** is a block diagram **800** illustrating a software architecture **802**, which can be installed on any one or more of the devices described herein. The software architecture **802** is supported by

hardware such as a machine **804** that includes processors **806**, memory **808**, and I/O components **810**. In this example, the software architecture **802** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **802** includes layers such as an operating system **812**, libraries **814**, frameworks **816**, and applications **818**. Operationally, the applications **818** invoke API calls **820** through the software stack and receive messages **822** in response to the API calls **820**.

(167) The operating system **812** manages hardware resources and provides common services. The operating system **812** includes, for example, a kernel **824**, services **826**, and drivers **828**. The kernel **824** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **824** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionalities. The services **826** can provide other common services for the other software layers. The drivers **828** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **828** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

(168) The libraries **814** provide a common low-level infrastructure used by the applications **818**. The libraries **814** can include system libraries **830** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **814** can include API libraries **832** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **814** can also include a wide variety of other libraries **834** to provide many other APIs to the applications **818**.

(169) The frameworks **816** provide a common high-level infrastructure that is used by the applications **818**. For example, the frameworks **816** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **816** can provide a broad spectrum of other APIs that can be used by the applications **818**, some of which may be specific to a particular operating system or platform.

(170) In an example, the applications **818** may include a home application **836**, a contacts application **838**, a browser application **840**, a book reader application **842**, a location application **844**, a media application **846**, a messaging application **848**, a game application **850**, and a broad assortment of other applications such as a third-party application **852**. The applications **818** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **818**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **852** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **852** can invoke the API calls **820** provided by the operating system **812** to facilitate functionalities described herein.

Additional Examples Include

(171) Example 1 is a method comprising: providing, by at least one processor, an XR user interface

of an XR system, the XR user interface comprising a virtual object displayed to a user; detecting, by the at least one processor, a first selection location of a pinch gesture being made by the user; detecting, by the at least one processor, a translation of the first selection location to a second selection location; generating, by the at least one processor, a rotated virtual object by rotating the virtual object using the first selection location and the second selection location; and displaying, by the at least one processor, the rotated virtual object to the user in the XR user interface.

(172) In Example 2, the subject matter of Example 1 includes, wherein detecting a selection location comprises: capturing, by the at least one processor, using one or more cameras of the XR system, tracking image data; determining, by the at least one processor, hand-tracking data using the tracking image data; and determining, by the at least one processor, the selection location using the hand-tracking data.

(173) In Example 3, the subject matter of any of Examples 1 to 2 includes, wherein determining the first selection location using the hand-tracking data comprises: generating, by the at least one processor, a rotate collider of the virtual object; generating, by the at least one processor, a selection location collider using skeletal model data of the hand-tracking data; and detecting, by the at least one processors, a collision between the selection location collider and the rotate collider.

(174) In Example 4, the subject matter of any of Examples 1-3 includes, wherein rotating the virtual object comprises: generating a first selection location vector using the first selection location and a virtual object center point of the virtual object; generating a second selection location vector using the second selection location and the virtual object center point of the virtual object; and rotating the virtual object using the first selection location vector and the second selection location vector.

(175) In Example 5, the subject matter of any of Examples 1-4 includes, rotating the virtual object using the first selection location vector and the second selection location vector comprises: determining a rotation angle using the first selection location vector and the second selection location vector; and rotating the virtual object using the rotation angle and an orientation vector of the virtual object.

(176) In Example 6, the subject matter of any of Examples 1-5 includes, detecting, by the at least one processor, that the pinch gesture is still being held by the user after redisplaying the rotated virtual object; in response to detecting that the pinch gesture is still being held, continuing, by the at least one processor, to rotate the virtual object using the pinch gesture.

(177) In Example 7, the subject matter of any of Examples 1-6 includes, wherein the XR system comprises a head-wearable apparatus.

(178) Example 8 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement of any of Examples 1-7.

(179) Example 9 is an apparatus comprising means to implement of any of Examples 1-7.

(180) Example 10 is a system to implement of any of Examples 1-7.

(181) Example 11 is a method to implement of any of Examples 1-7.

Conclusion

(182) Changes and modifications may be made to the disclosed examples without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

Glossary

(183) “Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

(184) “Client device” refers to any machine that interfaces to a communications network to obtain

resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

(185) “Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network, and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth-generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

(186) “Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processors. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g.,

hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

(187) “Machine-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “computer-readable medium,” “machine-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

(188) “Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-

storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

(189) “Non-transitory machine-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

(190) “Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

(191) Changes and modifications may be made to the disclosed examples without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

Claims

1. A method comprising: providing, by at least one processor, an XR user interface of an XR system, the XR user interface comprising a virtual object displayed to a user; providing, by the at least one processor, a plurality of rotate colliders corresponding respectively to an X axis, a Y axis, and a Z axis of the virtual object; determining, by the at least one processor, an axis to rotate the virtual object around based on a user selection of a rotate collider of the plurality of rotate colliders; detecting, by the at least one processor, a first selection location of a pinch gesture being made by the user; detecting, by the at least one processor, a translation of the first selection location to a second selection location; generating, by the at least one processor, a rotated virtual object by rotating the virtual object around the axis selected by the user using the first selection location and the second selection location; and displaying, by the at least one processor, the rotated virtual object to the user in the XR user interface.
2. The method of claim 1, wherein detecting a selection location comprises: capturing, by the at least one processor, using one or more cameras of the XR system, tracking image data; determining, by the at least one processor, hand-tracking data using the tracking image data; and determining, by the at least one processor, the selection location using the hand-tracking data.
3. The method of claim 2, wherein determining the first selection location using the hand-tracking data comprises: generating, by the at least one processor, a rotate collider of the virtual object; generating, by the at least one processor, a selection location collider using skeletal model data of the hand-tracking data; and detecting, by the at least one processor, a collision between the selection location collider and the rotate collider.
4. The method of claim 1, wherein rotating the virtual object comprises: generating a first selection location vector using the first selection location and a virtual object center point of the virtual object; generating a second selection location vector using the second selection location and the virtual object center point of the virtual object; and rotating the virtual object using the first selection location vector and the second selection location vector.
5. The method of claim 4, rotating the virtual object using the first selection location vector and the

second selection location vector comprises: determining a rotation angle using the first selection location vector and the second selection location vector; and rotating the virtual object using the rotation angle and an orientation vector of the virtual object.

6. The method of claim 1, further comprising: detecting, by the at least one processor, that the pinch gesture is still being held by the user after redisplaying the rotated virtual object; in response to detecting that the pinch gesture is still being held, continuing, by the at least one processor, to rotate the virtual object using the pinch gesture.

7. The method of claim 1, wherein the XR system comprises a head-wearable apparatus.

8. A machine comprising: at least one processor; and at least one memory storing instructions that, when executed by the at least one processor, cause the machine to perform operations comprising: providing an XR user interface of an XR system, the XR user interface comprising a virtual object displayed to a user; providing a plurality of rotate colliders corresponding respectively to an X axis, a Y axis, and a Z axis of the virtual object; determining an axis to rotate the virtual object around based on a user selection of a rotate collider of the plurality of rotate colliders; detecting a first selection location of a pinch gesture being made by the user; detecting a translation of the first selection location to a second selection location; generating a rotated virtual object by rotating the virtual object around the axis selected by the user using the first selection location and the second selection location; and displaying the rotated virtual object to the user in the XR user interface.

9. The machine of claim 8, wherein detecting a selection location comprises: capturing using one or more cameras of the XR system, tracking image data; determining hand-tracking data using the tracking image data; and determining the selection location using the hand-tracking data.

10. The machine of claim 9, wherein determining the first selection location using the hand-tracking data comprises: generating a rotate collider of the virtual object; generating a selection location collider using skeletal model data of the hand-tracking data; and detecting a collision between the selection location collider and the rotate collider.

11. The machine of claim 8, wherein rotating the virtual object comprises: generating a first selection location vector using the first selection location and a virtual object center point of the virtual object; generating a second selection location vector using the second selection location and the virtual object center point of the virtual object; and rotating the virtual object using the first selection location vector and the second selection location vector.

12. The machine of claim 11, wherein rotating the virtual object using the first selection location vector and the second selection location vector comprises: determining a rotation angle using the first selection location vector and the second selection location vector; and rotating the virtual object using the rotation angle and an orientation vector of the virtual object.

13. The machine of claim 8, wherein the operations further comprise: detecting, by the at least one processor, that the pinch gesture is still being held by the user after redisplaying the rotated virtual object; in response to detecting that the pinch gesture is still being held, continuing, by the at least one processor, to rotate the virtual object using the pinch gesture.

14. The machine of claim 8, wherein the XR system comprises a head-wearable apparatus.

15. A machine-storage medium including instructions that, when executed by a machine, cause the machine to perform operations comprising: providing, by at least one processor, an XR user interface of an XR system, the XR user interface comprising a virtual object displayed to a user; providing a plurality of rotate colliders corresponding respectively to an X axis, a Y axis, and a Z axis of the virtual object; determining an axis to rotate the virtual object around based on a user selection of a rotate collider of the plurality of rotate colliders; detecting, by the at least one processor, a first selection location of a pinch gesture being made by the user; detecting, by the at least one processor, a translation of the first selection location to a second selection location; generating a rotated virtual object by rotating the virtual object around the axis selected by the user using the first selection location and the second selection location; and displaying, by the at least one processor, the rotated virtual object to the user in the XR user interface.

16. The machine-storage medium of claim 15, wherein detecting a selection location comprises: capturing, by the at least one processor, using one or more cameras of the XR system, tracking image data; determining, by the at least one processor, hand-tracking data using the tracking image data; and determining, by the at least one processor, the selection location using the hand-tracking data.
17. The machine-storage medium of claim 16, wherein determining the first selection location using the hand-tracking data comprises: generating, by the at least one processor, a rotate collider of the virtual object; generating, by the at least one processor, a selection location collider using skeletal model data of the hand-tracking data; and detecting, by the at least one processors, a collision between the selection location collider and the rotate collider.
18. The machine-storage medium of claim 15, wherein rotating the virtual object comprises: generating a first selection location vector using the first selection location and a virtual object center point of the virtual object; generating a second selection location vector using the second selection location and the virtual object center point of the virtual object; and rotating the virtual object using the first selection location vector and the second selection location vector.
19. The machine-storage medium of claim 18, wherein rotating the virtual object using the first selection location vector and the second selection location vector comprises: determining a rotation angle using the first selection location vector and the second selection location vector; and rotating the virtual object using the rotation angle and an orientation vector of the virtual object.
20. The machine-storage medium of claim 15, wherein the XR system comprises a head-wearable apparatus.
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